

DIVIT RAWAL

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Research Interests. Statistical learning theory, high-dimensional and nonparametric statistics, learning under geometric structure, training dynamics and optimization in deep neural networks.

EDUCATION

University of California, Berkeley

2023 – 2027

B.A. in Statistics, Computer Science, and Physics

Berkeley, CA

Graduate-Level Coursework: Statistical Learning Theory; Theoretical Statistics; Mathematical Statistics; Learning for Dynamics and Control; Randomized Linear Algebra, Optimization, and Large-Scale Learning.

Teaching: EE 122 (Communication Networks) Head TA (Sp26), Reader (Sp25). Developed lab curriculum, wrote and graded exams and homeworks, and led weekly recitation sections.

PUBLICATIONS AND PREPRINTS

*($\alpha\beta$) denotes alphabetical ordering; * denotes equal contribution.*

[1] **Majority-of-Three is Optimal**

Divit Rawal, Nikita Zhivotovskiy. ($\alpha\beta$)

Under review, TMLR. [\[arXiv\]](#)

[2] **A Theory of Saddle Escape in Deep Nonlinear Networks**

Divit Rawal, Michael R. DeWeese.

Under review, NeurIPS 2026. [\[arXiv\]](#)

[3] **Rao-Blackwellized Score Matching on Manifolds**

Divit Rawal.

ICML 2026, Workshop on Structured Probabilistic Inference and Generative Modeling. [\[arXiv\]](#)

[4] **Minimax Rates for Hyperbolic Hierarchical Learning**

Divit Rawal, Sriram Vishwanath.

Preprint. [\[arXiv\]](#)

[5] **ALPHANSO: Open-Source Modeling of (α, n) Neutron Source Terms**

Divit Rawal, Anthony J. Nelson, William Zywiec, Daniel Siefman.

Nuclear Instruments and Methods in Physics Research, Section A. [\[doi\]](#)

Also presented at ANS Student Conference 2026 and INMM Annual Meeting 2026.

AWARDS AND GRANTS

Best Paper Award, American Nuclear Society Student Conference

Apr 2026

Best Paper in Mathematics, Computation, and AI Applications for “ALPHANSO: Open-Source Modeling of (α, n) Source Terms.”

VESSL AI Academia Grant

Aug 2025

Awarded for independent research into embeddings and information geometry; sole undergraduate recipient among cohort of Ph.D. students and postdoctoral researchers. (Supported work on [3] and [4].)

RESEARCH EXPERIENCE

Berkeley Statistical Machine Learning Group

Apr 2026 – Present

Researcher (advisor: Nikita Zhivotovskiy)

Berkeley, CA

Resolved open conjecture establishing the majority vote of three consistent classifiers as an optimal PAC learner in the realizable setting; authored resulting paper [1].

Developing new characterizations under which proper learning algorithms are optimal, tightening previous bounds, with applications to sample compression schemes.

Berkeley Artificial Intelligence Research (BAIR)

Sep 2024 – Present

Researcher (advisor: Michael DeWeese)*Berkeley, CA*

Derived layer-imbalance identity for nonlinear networks, extending deep-linear network analysis and predicting feature acquisition times in small-initialization neural networks; authored resulting paper [2].

Proved existence of in-context learning behavior in kernel ridge regression and analyzed emergence via spectral and random matrix theory methods.

Lawrence Livermore National Laboratory

Mar 2025 – Present

Researcher, Nuclear Science and Security Consortium Fellow (advisor: Daniel Siefman)*Livermore, CA*

Developed ALPHANSO [5], an open-source framework for deterministic modeling of (α, n) neutron production using continuous slowing-down approximations and modern evaluated nuclear data.

Developing new capabilities for FREYA event-by-event fission model, improving simulation of correlated neutron and gamma emission for radiation transport applications.

INVITED TALKS

ALPHANSO: Open-Source (α, n) Neutron Source Terms

Lawrence Livermore National Laboratory

Mar 2026

Lawrence Berkeley National Laboratory

Apr 2026

University of Tennessee, Knoxville

Jun 2026

INDUSTRY EXPERIENCE

Cisco Systems, Foundation AI

Jun – Aug 2025

*Software Engineer Intern**San Francisco, CA*

Developed components of the post-training and evaluation pipeline for cybersecurity-specialized LLMs based on Llama 3.1 8B, including instruction tuning and RLHF.

Contributed to the open-weight release of Foundation-Sec-8B and Foundation-Sec-8B-Instruct, including technical reports on data curation, training methodology, and benchmark evaluation.

ExperienceFlow AI

May – Sep 2024

*Machine Learning Engineer Intern**Remote*

Modeled finite state machine transition dynamics using transformer, state-space, and graph architectures; evaluated generalization and sample efficiency in low-label sequence prediction settings.

Developed RL-based control policies (DQN, SARSA) for dynamic state reasoning systems, achieving comparable performance to previous methods with 90% fewer samples (50,000 to 5,000).

Amazon

Aug – Dec 2023

*Software Engineer Intern**Remote*

Diagnosed and resolved production pipeline failures affecting 1M+ users; implemented a k -means clustering module in Java for OpenSearch ml-commons.

TECHNICAL SKILLS

Programming: Proficient in Python, C, C++, Java, SQL; intermediate in MATLAB, FORTRAN, Julia, R.**ML/AI:** Proficient in PyTorch; intermediate in JAX.**Other:** Intermediate proficiency in Lean, Mathematica.